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classmate

Date _____

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Decision Tree Classification

Total no: of machines = 8

Yes = 4, No = 4

Entropy (S) = $-P_{\text{yes}} \log_2(p_{\text{yes}}) - P_{\text{no}} \log_2(p_{\text{no}})$
 where $p \rightarrow$ probability of (yes/no) respectively.

$$P_{\text{yes}} = \frac{4}{8} \Rightarrow 0.5, P_{\text{no}} = \frac{4}{8} = 0.5$$

$$\begin{aligned} 2) S &= -(0.5) \log_2(0.5) - 0.5 \log_2(0.5) \\ &\rightarrow \boxed{S = 1} \end{aligned}$$

Entropy calculation

Temp = High $\rightarrow$ Yes: 3, No: 0 $\rightarrow$ no disorder $\rightarrow S = 0$ (High)

Temp = medium $\rightarrow$ Yes = 1, No = 1 $\rightarrow$ Entropy (S) = 1
 (Medium)

Temp = Low $\rightarrow$ Yes = 0, No = 3 $\rightarrow$ Entropy (Low) = 0

Weighted Entropy $\Rightarrow \frac{3}{8}(0) + \frac{2}{8}(1) + \frac{3}{8}(0)$
 $= 0 + 0.25 + 0 = \underline{\underline{0.25}}$

Information gain for temperature

$$IG(S, A) = \text{Entropy}(S) - \text{Weighted Entropy}(S|A)$$

$$\Rightarrow 1 - 0.25 = \underline{\underline{0.75}}$$

Entropy for vibration, i) vibration = high

$$\text{Yes} = 2, \text{No} = 1$$

$$\begin{aligned} \text{Entropy}(\text{high}) &\Rightarrow -\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \left(\frac{1}{3} \right) \log_2 \left(\frac{1}{3} \right) \\ &= -(0.667)(-0.585) \end{aligned}$$

$$= -(0.333)(-1.585)$$

$$= 0.390 + 0.528 = \underline{\underline{0.918}}$$

ii) vibration $\Rightarrow$ medium

$$\text{Yes} = 1, \text{No} = 1$$

$$\text{Entropy}(\text{medium}) = 1$$

(iii) vibration = Low $\rightarrow$ Yes: 1, No: 2

$$\rightarrow -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right)$$

$$= -(0.333)(-1.585) - (0.667)(-0.585)$$

$$= 0.528 + 0.390$$

$$= \underline{\underline{0.918}}$$

Final entries $\rightarrow$ Vibration	Entropy
High	0.918
Medium	1
Low	0.918

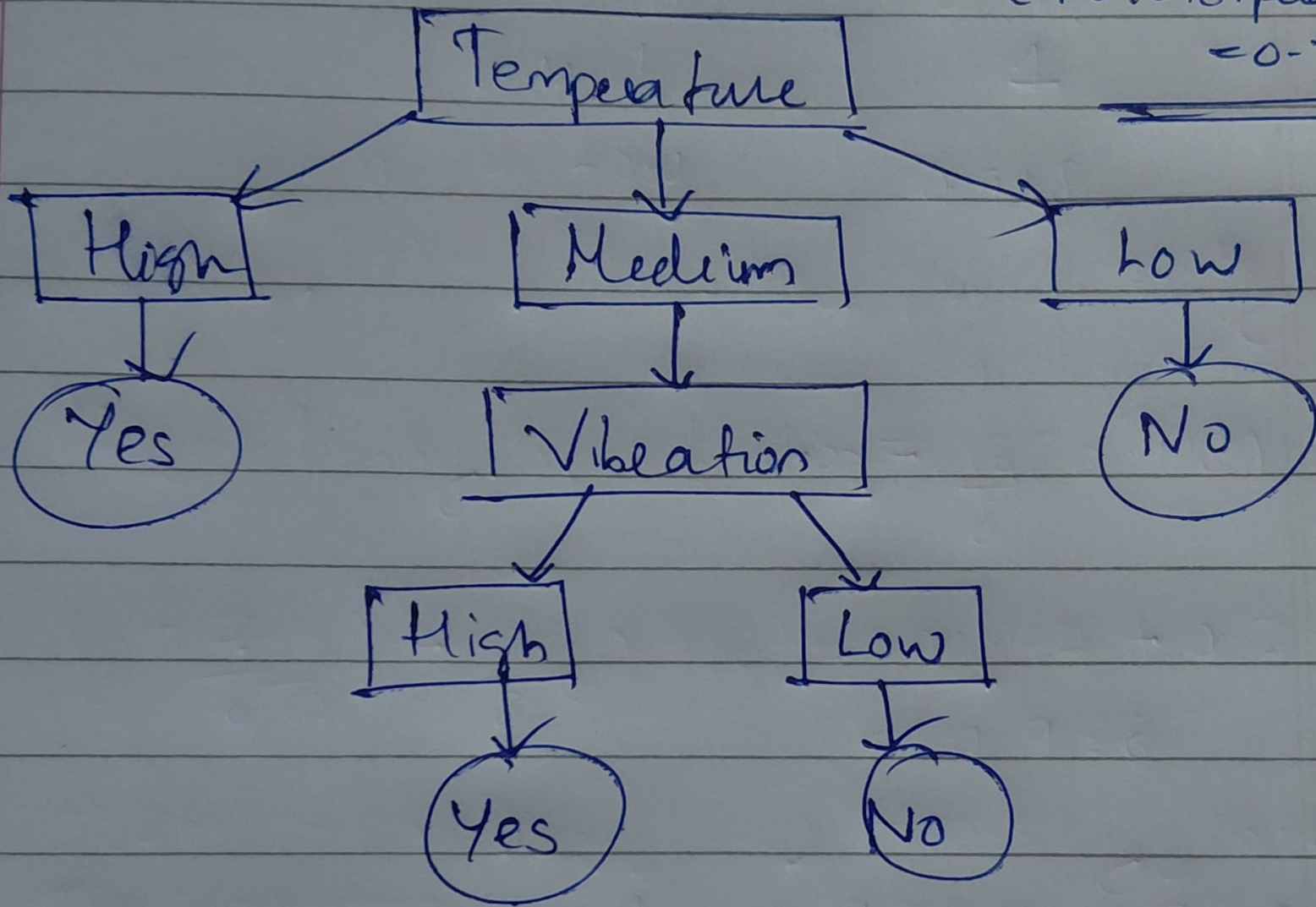
$$\text{Weighted entry for vibration} = \frac{3(0.918)}{8}$$

$$+ \frac{2(1)}{8} + \frac{3(0.918)}{8} = 0.344 + 0.25 + 0.344$$

$$= 0.938$$

$$IGI(\text{vibration}) = 1 - 0.938 = \underline{\underline{0.062}}$$

Decision tree → Root node → Temperature
 (For Temperature = 0-75)



Prediction for new machine

Temperature = High, Vibration = Medium
 path followed → Temp → High → Yes

Machine failure = Yes

predicted output → Failure